

User: ****Implement Task 2 from the README: "First to X Wins" match victory state.****

Add a configurable target win score, defaulting to 3. The game should continue playing rounds until either the player or CPU reaches the target score. When the target is reached, stop accepting further choice-button input, display a clear end-of-match victory banner showing whether the PLAYER or CPU won, and prompt the user to press R to restart the match. Reset both scores, choices, result state, and Game Over state when R is pressed. Preserve the existing Task 1 winner evaluation fix, the existing 1.8-second result display, and all existing UI behavior. Do not implement Task 3 or Task 4 yet. Make only the changes necessary for Task 2.

GitHub Copilot: I'll trace the README's Task 2 requirements into the game loop and button handling, then make the smallest targeted change and run the project's available checks.

Read memory

Search complete: Locate Task 2 code paths

Searched codebase for "README Task 2 First to X Wins game engine button input flow scoring", 4 results

Searched for files matching `\*\*/README\*`, 1 match

Searched for files matching `\*\*/game\_engine\*`, 2 matches

Searched for files matching `\*\*/\*button\*`, 2 matches

Searched for files matching `\*\*/\*score\*`, no matches

Searched for regex `First to X Wins|first to x wins`, 1 result

Searched for regex `scoring|score.\*update|score.\*increment`, 1 result

Searched for regex `delayed.\*result|result.\*display|banner.\*message`, 2 results

Searched for regex `choice.\*button|button.\*event|click.\*handler`, 8 results

Searched for regex `restart.\*handle|restart.\*game`, 1 result

Searched codebase for "First to X Wins scoring victory condition game over banner restart", 4 results

Searched for regex `First to X Wins|first to x wins|first to X wins`, 1 result

Searched for regex `target.\*win|win.\*ceiling|match.\*victory`, 2 results

Searched for regex `victory.\*banner|banner.\*victory|end.\*of.\*match`, 1 result

Searched for regex `self.player\_score|self.cpu\_score|player\_score|cpu\_score`, 6 results

Searched for regex `restart|Restart|start.\*over|play.\*again`, 3 results

Searched for regex `1800|display\_duration|result.\*visible|delayed.\*result`, 2 results

Searched for files matching `\*\*/\*victory\*`, no matches

Searched for files matching `\*\*/\*banner\*`, no matches

No matches found. Your search pattern might be excluded completely by either the search.exclude settings or .ignore files.

If you believe that it should have results, you can check into the .ignore files and the exclude setting (here are some excluded patterns for

reference:[**/node_modules,**/bower_components,**/*.code-search]).

Then if you want to include those files you can call the tool again by setting "includeIgnoredFiles" to true.

Searched codebase for "First to X Wins scoring victory condition game over banner restart press R", 4 results

Searched for text `First to X Wins`, 1 result

No matches found. Your search pattern might be excluded completely by either the search.exclude settings or .ignore files.

If you believe that it should have results, you can check into the .ignore files and the exclude setting (here are some excluded patterns for

reference:[**/node_modules,**/bower_components,**/*.code-search]).

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If you believe that it should have results, you can check into the `.ignore` files and the `exclude` setting (here are some excluded patterns for reference: `[**/node_modules,**/bower_components,**/*.code-search]`). Then if you want to include those files you can call the tool again by setting `"includeIgnoredFiles"` to `true`.

No matches found. Your search pattern might be excluded completely by either the `search.exclude` settings or `.ignore` files.

If you believe that it should have results, you can check into the `.ignore` files and the `exclude` setting (here are some excluded patterns for reference: `[**/node_modules,**/bower_components,**/*.code-search]`). Then if you want to include those files you can call the tool again by setting `"includeIgnoredFiles"` to `true`.

Searched for text ``restart after Game Over``, 1 result

Searched for text ``press R to restart``, 1 result

Searched for text ``end-of-match``, 1 result

Searched for text ``First to X``, 1 result

The controlling logic is in ``GameEngine``: rounds increment scores, mouse input directly starts another round, and ``update()`` clears each result after 1.8 seconds. I'm checking the full render/event code and project setup now; the key hypothesis is we can preserve the result delay while setting a terminal match state after scoring and gate further clicks until ``R``.

Read `[]()`

Read `[]()`

Read `[]()`

Read `[]()`

The implementation can stay in ``game_engine.py``: add ``target_score=3``, record the winner as soon as a score reaches it (which immediately blocks choice clicks), keep the final round's existing 1.8-second result visible, then switch that display area to a match-winner banner and restart prompt. ``R`` will reset the match state once the banner is shown. I'm applying that focused change now.

Apply Patch

Generating patch (109 lines) in `[]()`

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The match state and rendering changes are in. I'm doing a focused headless Pygame check now: verify configurable/default target behavior, both winner paths, blocked post-match clicks, the 1.8-second transition, and `R` reset.

```
Ran terminal command: $env:SDL_VIDEODRIVER='dummy'; python -c "import pygame; import
game.game_engine as module; from game.game_engine import GameEngine; pygame.init();
engine=GameEngine(560, 360); assert engine.target_score == 3; engine=GameEngine(560,
360, target_score=1); module.random.choice=lambda choices: 'SCISSORS';
engine.play_round('ROCK'); assert (engine.player_score, engine.cpu_score, engine.game_over,
engine.match_winner)==(1,0,True,'PLAYER');
engine.handle_event(pygame.event.Event(pygame.MOUSEBUTTONDOWN, button=1,
pos=engine.buttons[1].rect.center)); assert engine.player_choice=='ROCK' and
engine.player_score==1;
engine.round_resolved_time=pygame.time.get_ticks()-engine.display_duration; engine.update();
assert not engine.showing_result and engine.player_choice is None;
engine.render(pygame.Surface((560,360)));
engine.handle_event(pygame.event.Event(pygame.KEYDOWN, key=pygame.K_r)); assert
(engine.player_score, engine.cpu_score, engine.player_choice, engine.cpu_choice,
engine.game_over, engine.match_winner)==(0,0,None,None,False,None);
module.random.choice=lambda choices: 'ROCK'; engine.play_round('SCISSORS'); assert
(engine.cpu_score, engine.game_over, engine.match_winner)==(1,True,'CPU'); print('focused
match-state checks passed')"
```

The behavior check is blocked because this Windows environment has no `python` command available (it resolves to the Store alias). I'll check for the Python launcher next; if that's absent too, I'll use the workspace's available diagnostics and inspect the patch for structural issues.

Ran terminal command: Get-Command py -ErrorAction SilentlyContinue | Select-Object
-ExpandProperty Source

Checked [], no problems found

No Python launcher is installed; the editor reports no syntax or type diagnostics for the touched file. One edge case is worth aligning with the request: `R` should reset any time the match is over, including during the final 1.8-second round display, so I'm removing that extra key restriction and clearing the round timer as part of the reset.

Generating patch (14 lines) in []()

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Checked `[]()`, no problems found

The reset now works immediately after the target is reached, and it clears the timer, picks, scores, round message, and match winner. Static diagnostics are clean; I'm doing a final whitespace/diff check, since runtime testing is unavailable without Python.

Ran terminal command: `git diff --check`

Ran terminal command: `git diff -- game/game_engine.py`

The final diff is larger than expected, so I'm checking whether that's just line-ending noise before I call this done. ``git diff --check`` ran alongside it; I'll keep the delivered change confined to the actual Task 2 logic.

Read `[]()`, lines 1 to 80

Read `[]()`, lines 1 to 110

Read `[]()`, lines 110 to 260

Implemented Task 2 in ``game_engine.py``: the target defaults to 3 and is configurable, reaching it ends the match and blocks choice clicks, and the winner banner appears after the existing 1.8-second result display. Pressing ``R`` resets the scores, choices, result, timer, and game-over state. The Task 1 winner evaluation remains intact.

Editor diagnostics are clean. I couldn't run the Pygame behavior check because Python is unavailable in the environment; ``git diff --check`` also flags carriage returns on added lines.

Made changes.

User: > ****Implement Task 3 from the README: "Adaptive AI Pattern Tracking." Analyze the existing Rock Paper Scissors code and add lightweight player-pattern tracking. The CPU should track the player's previous choices and use that history to predict the player's next likely choice, then choose a move that counters the prediction. Include a clear fallback behavior when there is not enough history to make a prediction. Preserve all existing functionality from Tasks 1 and 2, including correct winner evaluation, First to X Wins, Game Over behavior, and R restart. Do not implement Task 4 yet. Make only the changes necessary for Task 3.****

2. Let the AI modify the code

When VS Code asks for permission to run a test command, you can click **Allow** if it is just running/testing the project.

When it finishes, **click Keep** to keep the changes.

GitHub Copilot: I'll build on the existing match state in `GameEngine` and check the README's Task 3 acceptance details first. I'll use a short recent history, counter the most frequent prior choice once there are enough samples, and preserve random CPU picks as the explicit cold-start fallback.

Read []()

Read [](), lines 35 to 85

The README's requirement maps directly to `play\_round()`, where CPU choice is currently random. I'll add a two-sample minimum and a five-throw rolling window: with a repeated most-common choice, the CPU counters that prediction; before enough history exists, it remains random. `R` will also clear that history so a restarted match starts fresh.

Generating patch (60 lines) in []()

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Checked [](), no problems found

Implemented Task 3 in `game\_engine.py`. The CPU uses the player's last five throws to predict the most frequent choice and select its counter. With fewer than two recorded throws, it falls back to a random move; ties between predictions are also resolved randomly. Pressing `R` clears the history along with the existing match state.

Editor diagnostics report no errors. I couldn't run a gameplay test because Python is unavailable in the environment.

Made changes.

User: Implement Task 4 from the README: "Procedural Gesture Icons & Reveal Animations."
Add procedural/drawn gesture icons for ROCK, PAPER, and SCISSORS without using external

image assets. Add a short reveal animation when the player and CPU choices are shown. Keep the existing Rock Paper Scissors gameplay, scoring, Task 1 winner fix, Task 2 First to X Wins behavior, Task 3 adaptive AI, and R restart behavior unchanged. The icons should be rendered programmatically using Pygame drawing primitives. Do not add external image files or dependencies. Make only the changes required for Task 4.

GitHub Copilot: I'll work in the existing renderer so the new visuals stay procedural and don't disturb match logic. First I'm checking the current choice layout and timing state, then I'll add small Pygame-drawn hand icons and a synchronized, decaying shake during their reveal.

Checked `render_choices()`, no problems found

Read `render_choices()`

Read `render_choices()`, lines 44 to 78

The current screen has room for two side-by-side choice displays between the scoreboard and result text. I'll preserve the existing score, outcome, and restart flow, adding only render-time icon drawing plus a 650 ms decaying shake timer set when each round begins.

Generating patch (273 lines) in

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Checked `render_choices()`, no problems found

The renderer and three primitive-drawn gestures are in place, and editor diagnostics are clean. I spotted a small layout collision between the heading and the top of the taller finger shapes, so I'm moving the headings up slightly before the final check.

Generating patch (9 lines) in `render_choices()`

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Checked `render_choices()`, no problems found

Implemented Task 4 in `game\_engine.py`: ROCK, PAPER, and SCISSORS are drawn with Pygame primitives, and both choices use a synchronized 650 ms shake reveal. Existing scoring, match, adaptive AI, and restart logic remain unchanged.

Editor diagnostics report no errors. I couldn't run the game because Python is unavailable in the environment.

Made changes.